

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image
image = cv2.imread('Sample.jpg')

# Convert BGR to RGB
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# -----
# Step 1: Simulate Low Light
# -----
low_light = (image * 0.25).astype(np.uint8)

# -----
# Step 2: Add Sensor Noise
# -----
noise = np.random.normal(0, 20, low_light.shape).astype(np.int16)

noisy = low_light.astype(np.int16) + noise
noisy = np.clip(noisy, 0, 255).astype(np.uint8)
```

```
# -----  
# Step 3: Simulate Blur  
# -----  
blurred = cv2.GaussianBlur(noisy, (7,7), 2)  
  
# -----  
# Step 4: Brightness Enhancement  
# -----  
gray = cv2.cvtColor(blurred, cv2.COLOR_RGB2GRAY)  
  
equalized = cv2.equalizeHist(gray)  
  
# -----  
# Display Results  
# -----  
titles = ['Original',  
          'Low Light',  
          'Sensor Noise',  
          'Blurred',  
          'Enhanced']  
  
images = [image,  
          low_light,  
          noisy,  
          blurred,  
          equalized]
```

```
plt.figure(figsize=(18,8))

for i in range(len(images)):
    plt.subplot(1,5,i+1)

    if len(images[i].shape)==2:
        plt.imshow(images[i], cmap='gray')
    else:
        plt.imshow(images[i])

    plt.title(titles[i])
    plt.axis('off')

plt.show()
```



